ClipSAT : Linear Algebra Vectors, Matrices, Eigenvalues & Least Squares

Unit 5.01: Vector Spaces & The Four Fundamental Subspaces Name: _____

Q1. Find lVertmathbf{u}rVert for mathbf{u}=-9,-12 .

[ ] A: 21
[ ] B: 14
[ ] C: 15
[ ] D: 225

Q2. A 3 times 3 symmetric matrix Q_{k} has eigenvalues lambda_1 = 12 , lambda_2 = 3 , and lambda_3 = 14 . Determine the trace text{tr}Q_{k} .

[ ] A: 26
[ ] B: 32
[ ] C: 29
[ ] D: 41

Q3. A linear transformation T: mathbb{R}^2 to mathbb{R}^2 is defined by Tmathbf{x} = Amathbf{x} , where A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . If mathbf{v} = begin{pmatrix} 1 2 end{pmatrix} , find Tmathbf{v} .

[ ] A: begin{pmatrix} 4 7 end{pmatrix}
[ ] B: begin{pmatrix} 3 7 end{pmatrix}
[ ] C: begin{pmatrix} 4 5 end{pmatrix}
[ ] D: begin{pmatrix} 5 7 end{pmatrix}

Q4. Consider the matrix A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . If a point P=x,y is transformed by A to P'=x',y' , such that begin{pmatrix} x' y' end{pmatrix} = A begin{pmatrix} x y end{pmatrix} , what is the image of the point 3,1 under this transformation?

[ ] A: 7, 6
[ ] B: 6, 7
[ ] C: 8, 5
[ ] D: 5, 8

Q5. Consider two vectors mathbf{v}_1 = begin{pmatrix} 3 4 end{pmatrix} and mathbf{v}_2 = begin{pmatrix} -4 3 end{pmatrix} . What is the dot product mathbf{v}_1 cdot mathbf{v}_2 ?

[ ] A: 1
[ ] B: 0
[ ] C: -7
[ ] D: 25

Q6. Consider the following system of linear equations represented by the augmented matrix: begin{pmatrix} 1 & 2 & | & 5 2 & -1 & | & 0 end{pmatrix} . Use row reduction to transform the matrix into row-echelon form and determine the unique solution x, y for the system. What is the value of x ?

[ ] A: 1
[ ] B: 2
[ ] C: 3
[ ] D: 4

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Q7. Consider the vector mathbf{v} = begin{pmatrix} 5 -12 end{pmatrix} . Calculate the magnitude of mathbf{v} , denoted as lVertmathbf{v}rVert .

[ ] A: 13
[ ] B: 17
[ ] C: 7
[ ] D: 169

Q8. Consider the vector mathbf{v} = begin{pmatrix} 3 4 end{pmatrix} . If this vector is plotted on a standard xy -plane, it represents a directed line segment from the origin to the point 3,4 . What is the geometric interpretation of the length of this vector?

[ ] A: The vector's direction in the xy-pla
[ ] B: The angle the vector makes with the
[ ] C: The distance of the point 3,4 from t
[ ] D: The sum of the components of the vec